

Estimating a Mundlak (1961) production function on the GLSS1–GLSS2 panel

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1 The short version

You can do it, on 506 households observed twice, and the binding constraints aren’t the ones you’d guess. Output is the weak variable, not labour: Section 9 never asks how much was harvested, only how much was sold, so the dependent variable has to be assembled from four disposition *values* and is missing entirely for about a fifth of cropped rows. Labour is in good shape once family and hired are added together. Land is the regressor the exercise is most likely to be about, and it’s the one carrying a measurement problem large enough to change the answer.

What follows is the recipe, then the diagnostics you should run before believing any of it.

2 What Mundlak's device actually is

Mundlak's 1961 paper asks how to estimate a production function when the firm knows something the econometrician doesn't. Write output for farm i in period t as

$$\ln Y_{it} = \alpha + \beta' \ln X_{it} + \mu_i + \varepsilon_{it},$$

where X_{it} collects the measured inputs and μ_i is management ability: fixed for the farm, unobserved, and—this is the whole problem— correlated with how much of everything the farmer chooses to apply. Pooled OLS on that equation gives you β contaminated by $\text{Cov}(\ln X, \mu)$, which is exactly the "management bias" of the title. Good managers apply more inputs and get more output; the input coefficients absorb the manager.

Mundlak's answer is to difference the manager away. With panel data you can estimate

$$\ln Y_{it} - \overline{\ln Y_i} = \beta'(\ln X_{it} - \overline{\ln X_i}) + (\varepsilon_{it} - \bar{\varepsilon}_i),$$

which is the within estimator, and μ_i is gone because it doesn't vary over t . The later and now more familiar formulation (Mundlak 1978) keeps the levels and adds the farm means as regressors,

$$\ln Y_{it} = \alpha + \beta' \ln X_{it} + \gamma' \overline{\ln X_i} + v_{it},$$

which recovers the same β while letting time-invariant covariates survive. The two are algebraically equivalent for β ; the correlated-random-effects form is more convenient when you want to test $\gamma = 0$, which is a test of whether the management bias was there at all.

Two things follow for our data, and both are unforgiving. With $T = 2$ the within estimator *is* first differences, so the whole exercise reduces to a cross-section of 506 differenced observations. And identification comes entirely from within-farm movement, which means any measurement error that moves across the two rounds goes straight into β .

3 The tables you need

All five ship in `lsms_library` 0.14.0 and cover 1987-88 and 1988-89 only. Agriculture is Section 9 in GLSS1/GLSS2 and Section 8 from GLSS3 on, and those two rounds are the only panel GhanaLSS has: the GLSS1 $\rightarrow$ GLSS2 rotating half, 712 households.

table	index	what you take from it
<code>crop_production</code>	<code>(t, i, plot_id, j)</code>	output values, <code>Area_ha</code>
<code>plot_inputs</code>	<code>(t, i, plot_id, input, j)</code>	<code>Cost</code> of purchased inputs
<code>plot_labor</code>	<code>(t, i, plot_id, stage, source, worker)</code>	hired <code>Cost</code> , exchange <code>PersonDays</code>
<code>labor</code>	<code>(t, i, pid, job)</code>	family hours, and the wage
<code>livestock</code>	<code>(t, i, animal)</code>	herd stock and flows

`plot_id` is 'Farm-level' everywhere in this country. Section 9 has no farm or plot roster at all, so the axis is kept and marked rather than dropped; see [GhanaLSS/_/CONTENTS.org](https://ghanalss.org/), "The Farm-level convention".

`j` is a real crop on the per-crop rows and 'Farm-level' on the expenses that aren't attributable to one.

4 Building the household-year panel

Everything below aggregates to (t, i) . The code is the working version; it's deliberately explicit rather than clever, because every one of these choices is a decision someone will want to revisit.

```
import numpy as np, pandas as pd, lsms_library as ll

C = ll.Country('GhanaLSS')

def hh(s, name):
    """Sum a served column to the household-year grain."""
    return s.groupby(level=['t', 'i']).sum(min_count=1).rename(name)

# ---- output: the four Section 9B dispositions -----
cp = C.crop_production()
out = hh(cp[['Value_sold', 'Value_seed', 'Value_given', 'Value_lost']]
        .sum(axis=1, min_count=1), 'output')
area = hh(cp['Area_ha'], 'area_ha')

# ---- the agricultural wage: a DAY rate, taken directly -----
lab = C.labor()
agri = lab['Industry'].between(100, 199)
wage_rows = lab[agri & (lab['OwnFarmOrBusiness'] == False)
               & (lab['Earnings'].fillna(0) > 0)]
day_wage = (wage_rows[wage_rows['EarningsUnit'].astype(str) == 'Day']
            ['Earnings'].groupby(level='t').median())
hours_per_day = lab.loc[agri, 'HoursPerDay'].dropna().median() # 5.0

# ---- labour, all three kinds, in person-days -----
annual_hours = lab['WeeksPerYear'] * lab['HoursPerWeek']
own = lab['OwnFarmOrBusiness'].fillna(False).astype(bool) & agri
family_days = hh(annual_hours.where(own), 'family_hours') / hours_per_day

pl = C.plot_labor()
source = pl.index.get_level_values('source')
hired_cost = hh(pl['Cost'].where(source == 'hired'), 'hired_cost')
exchange_days = hh(pl['PersonDays'].where(source == 'exchange'), 'exchange_days')

d = pd.concat([out, area, family_days.rename('family_days'),
              hired_cost, exchange_days], axis=1)
d['hired_days'] = d['hired_cost'] / d.index.get_level_values('t').map(day_wage)

# Sum the three. A household with no hired labour still has family labour,
# so the total is positive and log() survives -- this is what keeps the
# sample from collapsing.
parts = ['family_days', 'hired_days', 'exchange_days']
observed = d[parts].notna().any(axis=1)
d['labor_days'] = d[parts].fillna(0).sum(axis=1).where(observed)

# ---- purchased inputs -----
d['input_cost'] = hh(C.plot_inputs()['Cost'], 'input_cost')
```

5 Estimation

With two rounds, the within transformation and first differencing coincide, so the simplest correct thing is to difference. We write both forms because the correlated-random-effects version gives you the specification test for free.

```
cols = ['output', 'area_ha', 'labor_days'] # add 'input_cost' with care
panel = d[cols].dropna()
panel = panel[(panel > 0).all(axis=1)]
keep = panel.groupby(level='i').size().ge(2)
panel = panel[panel.index.get_level_values('i').isin(keep[keep].index)]
lg = np.log(panel)

# --- within / first differences: Mundlak's 1961 estimator -----
within = lg - lg.groupby(level='i').transform('mean')
y, X = within['output'], within[['area_ha', 'labor_days']]
beta = np.linalg.lstsq(X.values, y.values, rcond=None)[0]

# --- correlated random effects: levels plus farm means (Mundlak 1978) ---
means = lg.groupby(level='i').transform('mean')
design = pd.concat([lg[['area_ha', 'labor_days']],
                  means[['area_ha', 'labor_days']].add_suffix('_mean')],
                  axis=1)
design.insert(0, 'const', 1.0)
gamma = np.linalg.lstsq(design.values, lg['output'].values, rcond=None)[0]
```

The coefficients on the two `_mean` terms are Mundlak's γ . Testing them jointly zero is a test of whether management bias was present; if you can't reject, pooled OLS would have been fine and the panel bought you nothing. Report that test. It's the part of the exercise that says whether the exercise was needed.

Standard errors want clustering at the farm, and with $T = 2$ and 506 farms the asymptotics are thin enough that you should say so rather than lean on them.

6 The sample you actually get

specification	households	household-years
output, land, total labour	506	4,389
the same, plus purchased input cost	333	3,307

Within-household share of the log variance, which is what identifies β :

variable	share
output	0.229
area_ha	0.529
labor_days	0.195
input_cost	0.239

Adding family and hired labour into one input is what makes the first row possible. Zero of 4,842 household-years have `labor_days` equal to zero, because family labour is positive almost everywhere. Keeping hired labour as its own regressor instead costs 274 households, since a quarter of farms hire nobody and $\ln 0$ isn't defined.

7 Before you believe any of it

7.1 Land is the problem

`area_ha` has a within-household standard deviation of 1.12 in logs. A typical farm's reported cultivated area changes by a factor of about three between the two rounds. Land holdings don't do that.

The likely mechanism is definitional rather than real. Section 9 Part B Q2 asks acres harvested *per crop*, and the household total is a sum over the crops reported that year, so a farm reporting three crops in one round and five in the next shows a large "area" change that's partly a change in what got enumerated. A within estimator is precisely what this contaminates, and land is the coefficient you're most likely to care about.

Two cheap diagnostics, both worth running first:

- the distribution of crops-reported-per-household across the two rounds; and
- the area change restricted to farms reporting the *same crop set* in both.

If the second is much smaller than 1.12, you have your answer, and the fix is to estimate on the stable-crop-set subsample or to build area from a consistent crop basket.

7.2 Output is partial, and missing in a way that correlates with the thing you're studying

The four dispositions are sold, kept for seed, given away, and lost. They do not include what the household ate. Section 9 never asks it; Section 12B does, and the value is already derived (`GhanaLSS::food_acquired::12b-fortnight`), but joining the two needs a crop $\rightarrow$ food-item mapping and crosses a recall window.

Measured: about 19% of cropped rows carry no Section 9 value at all, covering roughly 14% of harvested acres and touching about half the farming households. Those are crops grown and wholly consumed. So the dependent variable is systematically thinner for subsistence-oriented farms, which makes it a *selected* measure rather than a noisy one, and the selection runs along exactly the market-orientation margin that a production function's error term would like to be independent of.

Do not patch this by imputing output from acreage. That's circular: acreage is a regressor, and you'd be manufacturing the coefficient you're trying to estimate.

7.3 Purchased inputs still carry a zero problem

`input_cost` is zero or missing for about a quarter of household-years, and that's what takes 506 farms down to 333. Adding it to labour isn't available to you—fertilizer and person-days don't add—so it needs its own treatment: a use indicator plus an intensity term, or a levels specification. A silent listwise drop changes the population under study, and the FULL row of the table above should be read as a different sample rather than a smaller one.

7.4 The wage is thin, and it is a day rate

Family labour arrives as hours and hired labour as cedis, so combining them needs a wage. Take the *day* rate, directly: median 250 cedis in 1987-88 and 300 in 1988-89, from 83 and 125

observations of agricultural workers who are not working on their own farm. Converting the month- and year-paid workers to an hourly basis instead pulls in salaried positions from a different labour market, understates the wage by 19–31%, and produces a suspiciously flat nominal series across two rounds of substantial inflation.

Note what this implies. Family labour valued at that wage comes to 1.25 and 1.39 times measured Section 9B output. That’s the familiar smallholder result—the shadow wage sits below the market wage—and it’s an upper bound, since output is partial. It also means any profit-function variant of this exercise will show negative profits, and you should decide in advance whether that’s a finding or a symptom.

8 What this design can’t do

Plot-level anything Section 9 has no farm or plot roster. `plot_id` is `'Farm-level'` throughout, which is a statement about the instrument, not a gap in the wiring.

Separating family from hired labour in the production function you can keep them separate, but you’ll pay 274 households for it and you’ll be estimating on farms that hire. Hired labour is *used by* 52.3% of household-years and is about 5% of labour days: widespread and small.

More than two rounds GLSS3 onward are independent cross-sections. There is no third wave of these households.

Currency comparisons across the two rounds without care GLSS1/GLSS2 are pre-2007 cedis. The library’s currency layer handles the declared monetary columns; check `lsms_library/currency.py` before assuming a column is converted.

9 References

Mundlak, Y. (1961). "Empirical Production Function Free of Management Bias." *Journal of Farm Economics* 43(1), 44–56.

Mundlak, Y. (1978). "On the Pooling of Time Series and Cross Section Data." *Econometrica* 46(1), 69–85.

10 Provenance

Every figure above is measured on the built tables, not the raw files, using `lsms_library` 0.14.0 and the GhanaLSS extracts staged on the workshop hub. The session 5 notebook rebuilds the panel and the estimates from scratch, so you can check any number here against your own run rather than taking it on trust.

The instrument’s own idiosyncrasies — which question produced which column, and where a variable means something other than its name — are documented in the library’s `GhanaLSS/_/CONTENTS.org`, which is the file to read before trusting any column.